

Bayesian imaging using Plug Play priors: when Langevin meets Tweedie

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Abstract

The article develops a theoretical framework for addressing inverse problems in imaging, which consist in reconstructing an image from degraded observations while accounting for uncertainty. The approach is Bayesian and relies on Plug-and-Play methods, which use a denoising algorithm as an implicit prior. This denoiser, often implemented as a neural network, is embedded in a Langevin-type (MCMC) scheme to sample from the posterior distribution. Tweedie's identity links the denoiser to the score of a smoothed prior, enabling its use within a Langevin scheme. The expected results are a stable reconstruction, the ability to estimate posterior means (MMSE), and to quantify uncertainty, with theoretical convergence guarantees.

1 Introduction

Imaging inverse problems are commonly modeled through the linear observation equation

$$y = Ax + n, \quad n \sim \mathcal{N}(0, \sigma^2 I_m), \quad (1)$$

where $x \in \mathbb{R}^d$ denotes the unknown image, $A : \mathbb{R}^d \rightarrow \mathbb{R}^m$ is a known forward operator (e.g. blur, masking, subsampling), and $y \in \mathbb{R}^m$ represents the measured data. In most practical imaging scenarios, the operator A is either non-invertible or severely ill-conditioned, which implies that direct inversion is unstable and tends to amplify measurement noise. To overcome these difficulties, additional regularity assumptions or prior knowledge about the unknown image must be incorporated into the reconstruction process. The Bayesian framework provides a principled way to achieve this by modeling prior information as a probability distribution, which acts as an implicit regularizer and mitigates the effects of noise and ill-posedness.

By Bayes' theorem, the posterior distribution of the unknown image given the observations is expressed as

$$\log(p(x|y)) = \log(p(y|x)) + \log(p(x)) - \log(p(y)), \quad (2)$$

where $p(y|x)$ is the likelihood induced by the observation model, $p(x)$ encodes prior knowledge about plausible images, and $p(y)$ is a normalization constant independent of x . The posterior distribution combines information from both the data and the prior, thereby enabling statistically grounded estimation procedures.

The MMSE solution is defined as

$$\hat{x}_{\text{MMSE}} = \arg \min_{u \in \mathbb{R}^d} \mathbb{E} [\|x - u\|_2^2 | y] = \mathbb{E}[x | y] = \int_{\mathbb{R}^d} \tilde{x} p(\tilde{x} | y) d\tilde{x},$$

which represents the conditional expectation of the unknown image given the observations. Although optimal in a mean-square sense, the computation of this estimator is generally intractable because it requires integration over a high-dimensional probability space. Consequently, numerical approximations and sampling-based methods are required in practice.

A natural route for posterior sampling is provided by Langevin dynamics. Under suitable regularity assumptions, the stochastic differential equation

$$dX_t = \nabla_x \log p(X_t | y) dt + \sqrt{2} dB_t = \nabla_x \log p(y | X_t) dt + \nabla_x \log p(X_t) dt + \sqrt{2} dB_t, \quad (3)$$

admits $p(x|y)$ as an invariant distribution. This means that, in the limit of infinite time, the trajectories of the diffusion process are distributed according to the desired posterior. In practice, however, continuous-time dynamics cannot be simulated exactly and must be approximated via discrete-time algorithms. The Unadjusted Langevin Algorithm (ULA), obtained through Euler–Maruyama discretization, reads

$$X_{k+1} = X_k + \delta \nabla_x \log p(y|X_k) + \delta \nabla_x \log p(X_k) + \sqrt{2\delta} Z_{k+1}, \quad Z_{k+1} \sim \mathcal{N}(0, I_d), \quad (4)$$

where $\delta > 0$ controls the trade-off between discretization bias and mixing speed. Small values of δ improve the approximation of the continuous dynamics but slow down convergence, whereas larger values accelerate exploration at the cost of increased bias. For the Gaussian likelihood induced by (1), the gradient of the data-fidelity term is explicitly given by

$$\nabla_x \log p(y|x) = -\frac{1}{\sigma^2} A^\top (Ax - y), \quad (5)$$

which corresponds to the gradient of a quadratic least-squares cost. To incorporate prior information in a more flexible manner, one may replace the explicit gradient of $\log p(x)$ with a proximal or denoising operator. A common approach is to encode prior information via the smooth Moreau–Yoshida envelope, leading to the recursion

$$X_{k+1} = X_k + \delta \nabla \log p(y | X_k) + \frac{\delta}{\lambda} (\text{prox}_{\lambda U}(X_k) - X_k) + \sqrt{2\delta} Z_{k+1}, \quad (6)$$

where U is an implicit regularizer and $\text{prox}_{\lambda U}$ denotes its proximal operator. This formulation bridges optimization-based regularization and stochastic sampling, providing a mechanism to incorporate prior structure without explicitly specifying a probability density.

Plug-and-Play (PnP) methods were introduced to address the difficulty of modeling complex image priors through analytical expressions. Instead of defining $p(x)$ explicitly, PnP approaches replace the proximal mapping with a learned or pre-trained denoiser $D_\varepsilon : \mathbb{R}^d \rightarrow \mathbb{R}^d$. Such denoisers, often trained on large image datasets, act as data-driven regularizers and capture rich structural information that is difficult to model analytically. Originally developed within optimization algorithms, PnP techniques incorporate D_ε as a surrogate proximal operator in iterative schemes. More recently, these ideas have been extended to Bayesian computation, where denoisers are used to approximate score functions or gradient fields within Langevin-type recursions. However, a fundamental challenge arises: a generic denoiser does not necessarily correspond to the gradient of the log-density of any explicit prior distribution. Consequently, inserting D_ε into Langevin dynamics may break the theoretical guarantees of posterior sampling, and the resulting Markov chain might fail to target a well-defined invariant distribution. Establishing convergence and statistical consistency therefore requires additional assumptions on the denoiser, such as non-expansiveness or proximity to a gradient field.

2 Key Concept: From Moreau–Yoshida to Tweedie and PnP-Langevin Algorithms

The key contribution of the paper is to encode prior information using Tweedie’s identity, which links the score function of a smoothed density to an oracle denoising operator. In particular, for a Gaussian kernel of variance ε , Tweedie’s formula reads

$$\varepsilon \nabla \log p_\varepsilon(x) = D_\varepsilon(x) - x, \quad (7)$$

where p_ε denotes the convolution of the original image prior p with a Gaussian of variance ε . This identity establishes a direct connection between score-based generative modeling and optimal Bayesian denoising. The operator D_ε corresponds to the oracle MMSE denoiser, defined by

$$D_\varepsilon(x) = (2\pi\varepsilon)^{-d/2} \int_{\mathbb{R}^d} \tilde{x} \exp\left(-\frac{\|x - \tilde{x}\|^2}{2\varepsilon}\right) p(\tilde{x}) d\tilde{x}. \quad (8)$$

This expression represents the conditional expectation of the clean image given a noisy observation under additive Gaussian perturbations, and therefore minimizes the mean-square reconstruction error. In practice, the true MMSE denoiser is unavailable because the prior distribution p is unknown; instead, one relies on learned denoisers that approximate this optimal mapping from noisy to clean images.

Two algorithms are employed to target the MMSE estimator: PnP-ULA and PPnP-ULA, the latter being a projected and stabilized variant of the former. The PnP-ULA algorithm integrates the denoiser into a Langevin sampling scheme as follows: where C is a large convex compact set

Algorithm 1 PnP-ULA

Require: $N \in \mathbb{N}$, $y \in \mathbb{R}^m$, $\varepsilon, \lambda, \alpha, \delta > 0$, $C \subset \mathbb{R}^d$ convex and compact

Ensure: $2\lambda(2L_y + \alpha L/\varepsilon) \leq 1$ and $\delta < \frac{1}{3} (L_y + \frac{1}{\lambda} + \frac{\alpha L}{\varepsilon})^{-1}$

- 1: Initialization: choose $X_0 \in \mathbb{R}^d$, set $k = 0$
 - 2: **for** $k = 0$ to N **do**
 - 3: $Z_{k+1} \sim \mathcal{N}(0, I_d)$
 - 4: $X_{k+1} \leftarrow X_k + \delta \nabla \log p(y | X_k) + \frac{\alpha \delta}{\varepsilon} (D_\varepsilon(X_k) - X_k)$
 - 5: $+ \frac{\delta}{\lambda} (\Pi_C(X_k) - X_k) + \sqrt{2\delta} Z_{k+1}$
 - 6: **end for**
 - 7: **return** $\{X_k : k \in \{0, \dots, N+1\}\}$
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and Π_C denotes the projection onto C . The projection term prevents the iterates from diverging in regions where the Langevin dynamics might otherwise explore low-probability tails. This stabilization improves numerical robustness and ensures that the Markov chain remains within a bounded domain. The parameter $\alpha > 0$ controls the influence of the denoiser relative to the data fidelity term, thereby governing the trade-off between prior regularization and adherence to the observation model. Smaller values of α yield solutions closer to the maximum likelihood estimate, whereas larger values emphasize the structural prior encoded by the denoiser.

The PPnP-ULA algorithm further projects each iteration onto the constraint set C , yielding By construction, the iterates remain in C , which relaxes the stability constraints on the step

Algorithm 2 PPnP-ULA

Require: $N \in \mathbb{N}$, $y \in \mathbb{R}^m$, $\varepsilon, \lambda, \alpha, \delta > 0$, $C \subset \mathbb{R}^d$ convex and compact

- 1: Initialization: choose $X_0 \in C$, set $k = 0$
 - 2: **for** $k = 0$ to N **do**
 - 3: $Z_{k+1} \sim \mathcal{N}(0, I_d)$
 - 4: $X_{k+1} \leftarrow \Pi_C \left(X_k + \delta \nabla \log p(y | X_k) + \frac{\alpha \delta}{\varepsilon} (D_\varepsilon(X_k) - X_k) + \sqrt{2\delta} Z_{k+1} \right)$
 - 5: **end for**
 - 6: **return** $\{X_k : k \in \{0, \dots, N+1\}\}$
-

size and allows for larger values of δ . This typically results in faster mixing and improved computational efficiency, although at the cost of a potential increase in asymptotic bias due to the projection operation. The two schemes therefore exhibit complementary properties: PnP-ULA provides stronger theoretical guarantees under small step sizes, while PPnP-ULA enables greater flexibility and practical performance in high-dimensional imaging tasks.

3 Simulations

A simulation study was conducted on a deblurring inverse problem. We consider a linear degradation model of the form

$$y = Ax + \eta,$$

where A is a 5×5 box blur operator and $\eta \sim \mathcal{N}(0, \sigma^2 I)$ denotes additive white Gaussian noise. The experiments were performed on a noisy version of the well-known *Camerman* image, where the noise variance was set to $\sigma^2 = 1/255$. To approximate the MMSE estimator, we employed the

PPnP-ULA algorithm with a pre-trained denoiser from the `deepinv` library. The chosen denoiser is DnCNN. The algorithm parameters were set as follows: the regularisation term α is set to 1 for simplicity purposes, the denoising strength is $\varepsilon = 4/255^2$ and the total number of iterations $N = 4000$. For the step size δ , the condition for the convergence of the PPnP-ULA is $\delta < \frac{1}{\frac{L}{\varepsilon} + L_y}$.

In this case, we have $L = 1$ and $L_y = \frac{\|A^\top A\|}{\sigma^2}$. Since we have no access to the norm of this operator, we use the power iteration method in order to target the value of L_y . Then we chose to have as a step size $\delta = 0.5 \frac{1}{\frac{L}{\varepsilon} + L_y}$. The constraint set was defined as $C = [-1, 2]^d$. The Markov chain was initialized at X_0 , corresponding to the observed noisy image. A burn-in period of 500 iterations was discarded before computing empirical averages. To evaluate the reconstruction performance, we report the Structural Similarity Index Measure (SSIM) and the Peak Signal-to-Noise Ratio (PSNR).



Figure 1: Blurred Image



Figure 2: Unblurred Image

Applying the PPnP-ULA algorithm to Figure 1 yields the reconstruction shown in Figure 2. The result clearly falls short of expectations. One fundamental issue is that the use of a pre-trained neural network makes it difficult to ensure that the theoretical assumptions required for convergence—particularly the conditions imposed on the denoiser—are satisfied. Moreover, as with many Markov Chain Monte Carlo (MCMC) methods, the underlying Markov chain may require a large number of iterations before approaching a satisfactory estimator.

The paper provides limited detail regarding the selection and configuration of the denoiser used to obtain the reported results, which further complicates reproducibility and interpretation. Additionally, DnCNN is known to exhibit hallucination artifacts, which may explain the noticeable alterations in the background of the reconstructed image.

Using the chosen evaluation metric, we obtain:

4 Conclusion

This work addresses an imaging inverse problem in the context of image deblurring. The degradation process is modeled as a linear forward operator combined with additive Gaussian noise, leading to an ill-posed reconstruction problem in which direct inversion is unstable and highly sensitive to noise. To regularize the problem in a principled manner, we adopt a Bayesian framework that combines the likelihood induced by the observation model with prior information on the unknown image. Since the minimum mean square error (MMSE) estimator is defined as the posterior expectation, its computation requires integration over a high-dimensional distribution and is therefore intractable in practice. To approximate this estimator, we rely on Langevin dynamics, whose associated stochastic differential equation admits the posterior distribution as invariant

measure. A discretized version of this dynamics enables practical posterior sampling. To incorporate prior information without specifying an explicit analytical prior distribution, we rely on Tweedie’s identity, which links the score of a smoothed density to an oracle denoising operator. This connection allows us to encode prior information through a denoiser rather than through a closed-form prior. In practice, the ideal MMSE denoiser is unavailable, so we employ a pre-trained deep denoiser (DnCNN) as a surrogate prior model. The resulting algorithm is implemented within a Projected Plug-and-Play Unadjusted Langevin Algorithm (PPnP-ULA) framework. The projection step ensures numerical stability by constraining the iterates to a bounded convex set, while the denoiser replaces the explicit prior gradient in the Langevin recursion. In the deblurring simulation conducted on a noisy Cameraman image, the method produces a visually improved reconstruction compared to the blurred input. However, the result does not fully meet expectations. Artifacts are observed, particularly in the background, which may be attributed to the known hallucination behavior of the pre-trained denoiser. Furthermore, the theoretical convergence guarantees rely on assumptions on the denoiser that cannot be verified in practice. Although the approach demonstrates the feasibility of combining Bayesian sampling with learned priors, the experiment highlights the sensitivity of Plug-and-Play Langevin methods to denoiser properties, parameter tuning, and mixing behavior of the Markov chain.